

Care+

AI-Driven, Privacy-First Healthcare Companion for Elderly Home Care in Nepal

Innovation Project — Final Report

Module: Innovation Project (CMP4285)

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Table of Contents

Acknowledgement.....	2
Table of Contents.....	6
Executive Summary.....	7
Introduction.....	8
Background.....	8
Motivation.....	8
Why This Project Was Selected.....	8
Objectives.....	8
Problem Identification.....	10
Customer Discovery & Empathy.....	11
Target Users.....	11
Customer Interviews and Informal Feedback.....	11
User Personas.....	11
Customer Journey (Overview).....	12
Root Cause Analysis.....	13
Proposed Solution.....	14
System Overview.....	14
How It Works.....	14
Major Features.....	14
Value Proposition.....	14
MVP (Minimum Viable Product).....	16
Definition of the MVP.....	16
Core Features Developed.....	16
Features Intentionally Excluded.....	16

MVP Limitations.....	17
Future Improvements.....	17
Design Thinking Process.....	18
Empathize.....	18
Define.....	18
Ideate.....	18
Prototype.....	18
Test.....	18
Market & Competitor Analysis.....	19
Target Market.....	19
Competitor Comparison.....	19
Blue Ocean Opportunity.....	19
Unique Selling Proposition.....	20
Business Model.....	21
Revenue Model.....	21
Customer Value.....	21
Pricing Strategy.....	21
Sustainability.....	21
Business Model Canvas (Summary).....	21
SDG Alignment.....	23
Technology Stack.....	24
System Design.....	25
System Architecture.....	25
Database Design (ER Diagram).....	25
Use Case Diagram.....	25
System Flowchart.....	25

Wireframes / UI Design.....	25
AI Integration.....	26
Development Process.....	27
Team Responsibilities.....	27
Timeline / Milestones.....	27
Gantt Chart.....	27
Testing & Validation.....	29
Testing Methods.....	29
User Testing and Feedback.....	29
Bug Fixes and Improvements.....	29
Demonstration of MVP.....	30
Login.....	30
Dashboard.....	30
Features.....	30
AI Functionality.....	30
Final Output.....	30
Challenges & Solutions.....	31
Technical Challenges.....	31
Team Challenges.....	31
Time Management.....	31
Learning Experiences.....	31
Project Outcomes.....	32
Objectives Achieved.....	32
Customer Value Delivered.....	32
Skills Gained.....	32
Innovation Achieved.....	32

Future Scope.....	33
Conclusion.....	34
References.....	35
Appendices.....	36
Appendix A — Survey / Interview Questions.....	36
Appendix B — Raw Data.....	36
Appendix C — Additional Screenshots.....	36
Appendix D — Demonstration Video.....	36

Executive Summary

Care+ (CarePlus) is a privacy-focused, AI- and IoT-powered healthcare assistant that supports home-care provision for elderly people in Nepal. The project addresses the multi-layered problems of language barriers, poor medicine adherence monitoring, incomplete communication between caregivers, and low trust in cloud-based healthcare platforms, all of which affect elderly patients, caregivers, and family members living overseas. The proposed solution is a companion device based on the Raspberry Pi and a supporting mobile and web application ecosystem. Care+ uses voice-to-text conversion of speech via Whisper, sends the request to a FastAPI orchestration layer, which passes it to a locally-hosted language model, generates a response, and records the intake of medicine, mood, and care events that take place throughout the session. The system also provides near-real-time synchronization of role-based dashboards for doctors, guardians, caregivers, and administrators. The intended recipients are elderly self-reliant and semi self-reliant citizens, guardians and family members who are away from the country, home care nurses and small healthcare institutions. Development has been executed in an agile and milestones-driven methodology, broken down into the 4 focus areas: Core Communication and AI orchestration, Medical / functional Management, Emotional Analytics / Reporting, and Multi-user Ecosystem / Hardware integration. Medicines reminder scheduling, medicines intake notification, mood logging, caregiver dashboard synchronization and voice interaction are the core features of the MVP. The outcome of the Innovation Project is a working prototype, a clear go-to-market and revenue plan, and a clear roadmap leading up to the deployment of a pilot system in the Kathmandu Valley, making Care+ a locally relevant and scalable contribution to healthcare technology in Nepal for the elderly..

Introduction

Background

The ageing population of Nepal coupled with tremendous migration of the youths to foreign countries to study and work has created a constant dearth of ordinary elderly care within Nepal. Many seniors are living with reduced caregivers or without any caregiver at all and their adult children are tracking their parents' health remotely with regular phone calls. Today, the majority of digital health solutions adopted in Nepal are English language-based, cloud-based and conceptualized around hospital or telemedicine operational procedures, but not toward the holistic patient-centric care environment.

Motivation

In informal chats with families and carers, Team Git Force found that missing medicines and not responding quickly enough in emergency situations, and not knowing about an older person's emotional state, were common stressors. This motivated the team to create a companion that could function locally, communicate naturally in the Nepali language, and connect the elderly person with the wider care circle without requiring an internet connection or complex interfaces.

Why This Project Was Selected

The team chose this project as it has three areas of shared interest and academic relevance to the Innovation Project module: Applied Artificial Intelligence, Embedded/IoT Systems and Human-centred Healthcare Design. It also provided a realistic conception and scale for a team of four to develop an MVP that was both feasible within the length of the module and reflective of the SDGs.

Objectives

- Build a safe, local-first AI health companion for elderly home care.
- Remind and track adherence to medications.

- Offer role-specific access for physicians, guardians and caregivers.
- Generate interpretable emotional and medical trend reports.
- Ensure low-friction, Nepali-first voice interaction for daily use.

Problem Identification

In Nepal, home-based elderly care is hindered by language barriers, transport issues, and communication barriers, which together compromise the quality and safety of care. These issues involve several different groups, all of which have a similar root cause, but with each group encountering a different part of the issue.

User Group	Problem	Consequence
Elderly patients	Difficulty remembering medicine schedules and appointments	Missed doses and preventable health complications
Caregivers	Lack of centralized communication and monitoring tools	Poor coordination and delayed response to emergencies
Families abroad	Inability to monitor elderly relatives remotely	Anxiety and reduced trust in the quality of home care
Patients with language barriers	Healthcare applications not localized for Nepali users	Reduced accessibility and low technology adoption
Healthcare institutions	Limited real-time home monitoring systems	Delayed interventions and incomplete patient data

That's a big problem size is there are many elderly people in Nepal who don't have a full-time care taker in their household, and the current products available online that target the health of elderly people are only concerned with setting up hospital visits or measuring exercise but not language-matched, ongoing home care.

Customer Discovery & Empathy

Target Users

- Elderly individuals requiring medicine and daily care support at home.
- Family members or guardians living away from the patient, including abroad.
- Home-care nurses and caregivers managing multiple patients.
- Doctors who require concise adherence and mood trend reports.

Customer Interviews and Informal Feedback

With Care+, elderly consumers gain independence and reassurance through a user experience delivered in their native language, spoken words. It will provide transparency and assurance for guardians/caretakers and eliminate the need for manual monitoring. Recurring themes included difficulty remembering multiple medicine timings, discomfort with English-only mobile applications, and a desire for guardians abroad to receive simple, reassuring updates rather than raw data.

User Personas

Persona 1 — Maya Devi, 72, Retired, Kathmandu

Maya Devi owns a basic smartphone and prefers speaking over typing. She mostly struggles to remember when to take her medicines and is lost when it comes to English language applications. She would like to live independently in her home.

Persona 2 — Suman, 38, Software Engineer, Australia

Suman's parents are elderly residents of Nepal. He uses tech every day and his biggest worry is that his parents don't take their medicine on time. He wants to receive timely, reassuring updates without having to call every day.

Persona 3 — Nurse Rina, 29, Home-Care Worker, Lalitpur

Rina cares for a number of senior patients, and requires prompt and accurate health status reports. She currently manages repetitive checks and paper records, and hopes these processes become more streamlined so that she can provide more direct care.

Customer Journey (Overview)

Stage	User Action	Care+ Response
Discovery	Learns about Care+ through a healthcare provider	Demo and onboarding support
Setup	Installs the device and mobile application	Guided setup and language selection
Daily use	Receives medicine reminders	AI voice assistance and intake confirmation
Monitoring	Caregiver checks the dashboard	Real-time synchronization and reports
Long-term use	Builds trust in the system	Personalized recommendations and emotional trend tracking

Root Cause Analysis

To understand why home-based elderly care in Nepal remains fragmented, the team applied the 5 Why technique to the core symptom of missed medication and delayed caregiver response.

Level	Question	Answer
1	Why do elderly patients miss medicine doses?	They forget the schedule or are not reminded at the right time.
2	Why are they not reminded effectively?	Existing reminder tools are text-based, in English, or require manual interaction.
3	Why are these tools not usable for elderly Nepali speakers?	Most digital health products are designed for English-literate, tech-comfortable users.
4	Why has no localized alternative been adopted?	Building Nepali-language, voice-first AI systems requires local model orchestration, which few local products have invested in.
5	Why is this investment lacking?	The home-care segment has historically been underserved compared to hospital and fitness-tracking markets, reducing commercial incentive.

This analysis indicates that the root cause is structural: the underlying technology ecosystem in Nepal has prioritized institutional and fitness-oriented health products, leaving a gap in accessible, language-appropriate, home-based elderly care technology. Care+ is positioned to directly address this root cause rather than only its downstream symptoms.

Proposed Solution

System Overview

Care+ is a privacy-first, local AI and IoT healthcare companion that runs core speech and reasoning services on a Raspberry Pi edge device, paired with a companion mobile and web application. The system enables a complete, closed-loop care cycle rather than a single-purpose reminder tool.

How It Works

- The user speaks naturally in Nepali or simple English.
- Whisper speech-to-text converts the spoken input into text.
- A FastAPI orchestration layer routes the request to the local language model service.
- The language model determines intent and triggers the relevant action: conversational reply, medicine logic, object lookup, or reporting.
- The response is converted back to speech via text-to-speech and relevant care events are logged.
- Doctor, guardian, patient, and admin dashboards synchronize in near real time.

Major Features

- Nepali-first, multilingual voice interaction.
- Automated medicine reminders with intake confirmation tracking.
- Conversational AI assistant for general health-related queries.
- Mood and emotional trend logging derived from conversation patterns.
- Patient, guardian, doctor and administrator dashboards.
- Local-first processing to reduce dependence on continuous internet connectivity.

Value Proposition

At its core, Care+ is not telemedicine, hospital management, or a fitness tool—it is a home-based ecosystem that brings together local AI, IoT hardware, multilingual voice interaction, and emotional state monitoring. This integrated model is directly related to the root cause identified above: a lack of accessible, home-based healthcare technology in Nepal delivered in the Nepali language.

MVP (Minimum Viable Product)

Definition of the MVP

The Care+ MVP validates the core healthcare assistance loop: voice interaction, medicine reminder scheduling, medicine intake confirmation, mood logging, and caregiver dashboard synchronization, implemented on a Raspberry Pi prototype with connected mobile and web dashboards.

Core Features Developed

Feature	Description	Priority
Voice interaction	Nepali and English voice-based communication	Must have
Medicine reminder system	Automated medicine scheduling and reminders	Must have
Medicine intake logging	Confirmation tracking for medicine intake	Must have
AI chat assistant	Healthcare query support through local AI	Must have
Dashboard system	Patient and caregiver monitoring dashboards	Must have
Emotional assistant	Mood tracking based on conversation history	Should have
PDF health reports	Automated health report generation	Should have
Emergency alert system	Critical notifications sent to caregivers	Must have
Offline data synchronization	Local data storage with delayed sync	Nice to have

Features Intentionally Excluded

- Wearable device integration (heart rate, sleep, movement tracking).
- Advanced multilingual support beyond Nepali and English.
- Clinical integration APIs for hospital information systems.

- Fully autonomous offline operation without any periodic synchronization.

MVP Limitations

- Speech recognition accuracy may degrade in noisy home environments.
- Emotional analysis is based on conversational patterns and is indicative rather than clinically diagnostic.

Future Improvements

- Personalised risk alerts using longitudinal behavioural models.
- Wearable integration for physiological monitoring.
- Expanded language support for other South Asian languages.

Design Thinking Process

Empathize

The team conducted informal interviews with elderly individuals and caregivers in Nepal, as well as guardians living abroad, to understand their daily lives, the tools they currently use, the frustrations these tools cause, and the emotional challenges of providing care from a distance.

Define

Results were summarised in terms of a problem statement: Elderly speakers of Nepali cannot access a language-friendly, accessible means to manage their medicines and to communicate their health and wellbeing with a remote care circle.

Ideate

The team came up with several ideas: one that sends reminders via text, one that works entirely with wearables, and one that has an AI companion using voice inputs. Voice, with a local, independently hosted approach was chosen for its ease of use for non-English speaking, low-literacy and elderly users.

Prototype

A working prototype was built with a Raspberry Pi, Whisper speech to text, FastAPI orchestration, and a local language model (LLM), and using role-based dashboards for patients, guardians, doctors, and administrators over a web app and mobile app.

Test

The prototype was informally tested with some pilot users and internally by the team to assess the reliability of speech recognition, the success of the reminders, and usability of the dashboard.

Market & Competitor Analysis

Target Market

Segment	Characteristics	Opportunity
Elderly individuals	Require medication and emotional support	Growing ageing population
Families abroad	Need remote monitoring solutions	High emotional and financial value
Healthcare institutions	Need patient monitoring systems	Institutional adoption potential
Caregivers	Require communication and tracking tools	Daily operational support
Rural healthcare program	Limited healthcare access	Scalable social impact

Competitor Comparison

Competitor Type	Strength	Weakness
Generic telemedicine apps	Online consultations	No offline support or elderly focus
Smartwatch health trackers	Continuous health tracking	Expensive and internet-dependent
Hospital management software	Institutional management	Not designed for home-based care
International AI assistants	Advanced AI capabilities	No Nepali localization

Blue Ocean Opportunity

The majority of the digital health products on the market in Nepal are tools that allow booking appointments in hospitals, teleconsultation, or tracking fitness. There are not many that combine local AI processing, multilingual voice interaction, emotional monitoring and home care with an IoT solution.

Unique Selling Proposition

Care+'s USP is a privacy-first, Nepali-first AI companion that unifies medicine adherence, emotional monitoring, and multi-role caregiver coordination in a single home device, functioning reliably even where internet connectivity is inconsistent.

Business Model

Revenue Model

Proposed pricing includes a hardware/ setup fee and an ongoing monthly fee for reminder, dashboard and reporting services. In addition to sales to institutions for elderly care, premium tiers will be tailored for doctors for analytics, more advanced caregiver dashboards, and future wearable or hospital integration as well.

Customer Value

Care+ provides elderly users with independence and reassurance with an audio-visual experience in their native language. It delivers transparency and a sense of security for guardians and caregivers, and provides a scalable monitoring layer for home-based patients for institutions.

Pricing Strategy

Pricing is planned to be introduced gradually: subsidised pilot pricing during early deployment, followed by a standard hardware-plus-subscription model, and negotiated institutional licensing for healthcare partners at scale.

- Target Retail price (one time): Rs. 52,999 (excluding vat)
- Monthly subscription for cloud database sync: Rs. 2000 per month

Sustainability

Sustainability requires low per-unit hardware cost, growing subscription retention by proven health benefits, and building partnerships within the healthcare space, in addition to direct-to-consumer sales.

Business Model Canvas (Summary)

Component	Details
Key Partners	Elderly households, guardians abroad, home-care providers, clinics
Key Activities	AI development, healthcare monitoring, IoT integration
Value Proposition	Localized AI healthcare companion for elderly care
Customer Relationships	Subscription support and healthcare partnerships
Revenue Streams	Subscriptions, institutional licensing, premium features

SDG Alignment

SDG Goal	Care+ Contribution	Expected Impact
SDG 3 — Good Health and Well-Being	Supports preventive healthcare and medicine adherence	Improved elderly healthcare outcomes
SDG 9 — Industry, Innovation and Infrastructure	Introduces practical AI and IoT healthcare infrastructure	Increased healthcare technology adoption
SDG 10 — Reduced Inequalities	Improves healthcare accessibility for underserved and non-English-speaking communities	Better healthcare inclusion

SDG 3 was chosen as the principal goal because Care+ has a direct impact on medication adherence and on-going monitoring of wellbeing, the two areas most associated with impending health deterioration of elderly home care patients. SDG 9 and SDG 10 are supporting goals that reflect the project's technology-driven and inclusion-driven design choices respectively.

Technology Stack

Layer	Technology	Reason
Frontend	React Native / React-based dashboards	Cross-platform mobile and web development
Backend	FastAPI	High-performance AI request orchestration
Database	MongoDB / PostgreSQL	Storage of logs, medicines, and reports
AI model	Locally hosted LLM (e.g., Gemma via Ollama)	Local, privacy-preserving AI processing
Speech-to-text	Whisper	Accurate multilingual speech recognition
Hardware	Raspberry Pi with microphone and speaker modules	Affordable, privacy-first edge computing
Synchronization	WebSockets	Real-time dashboard updates
Deployment	Docker and cloud hosting	Scalable, containerized deployment

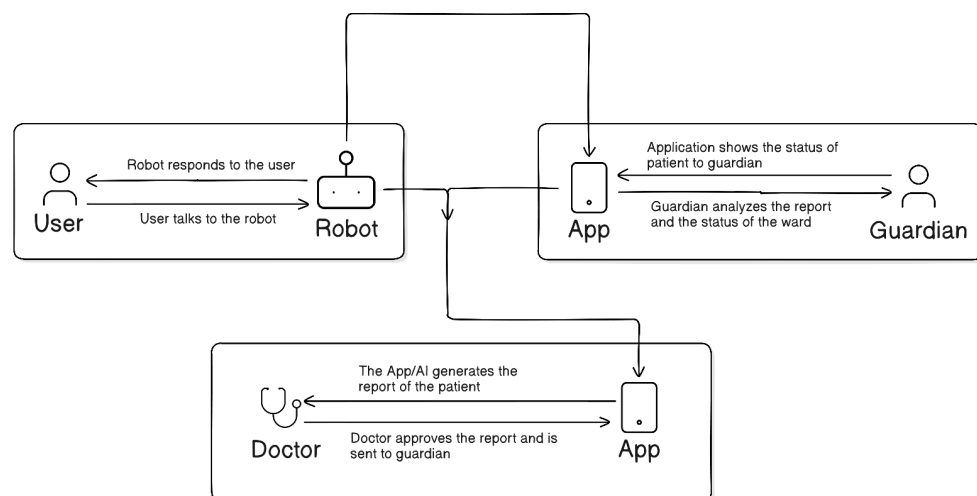
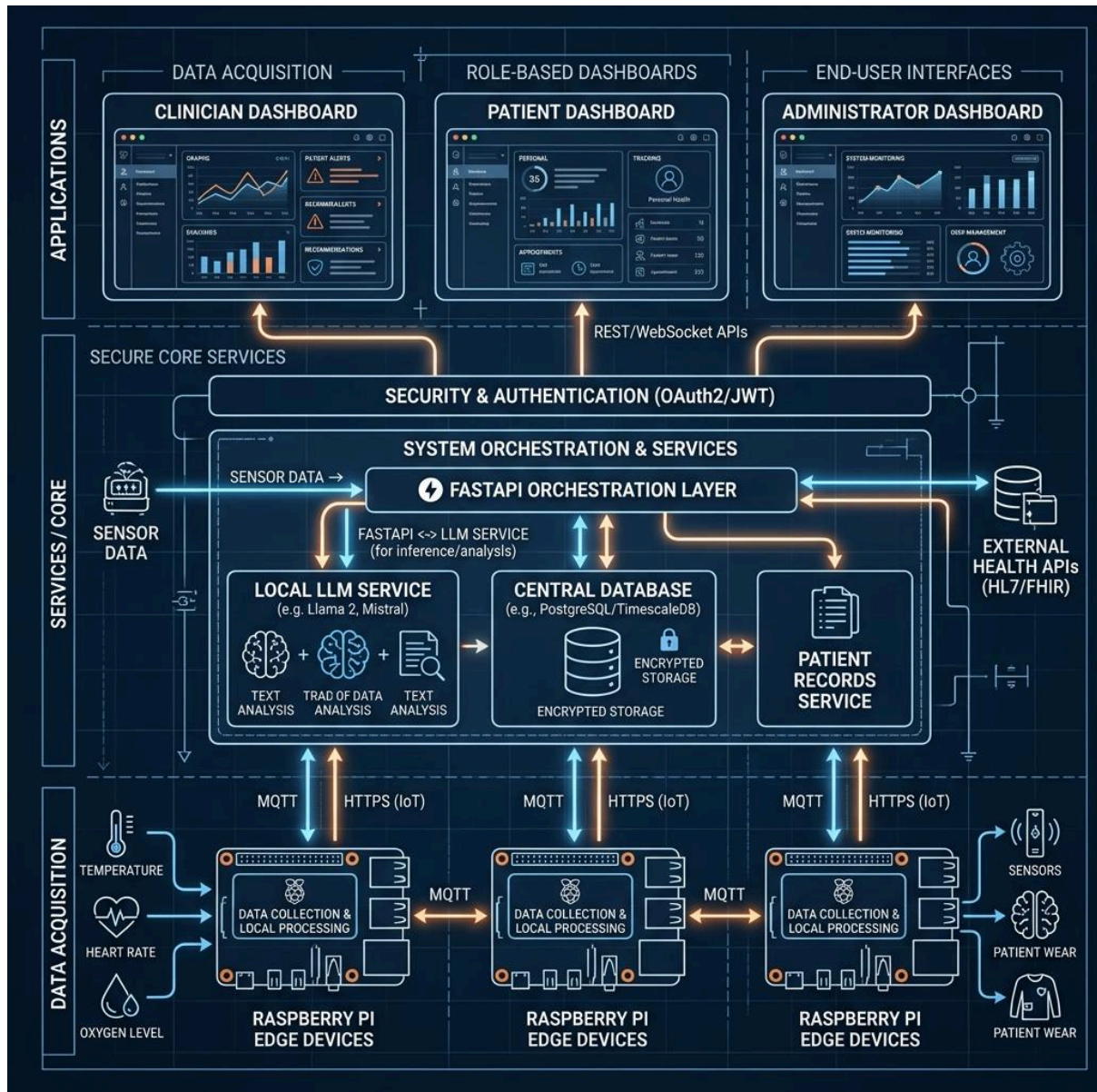


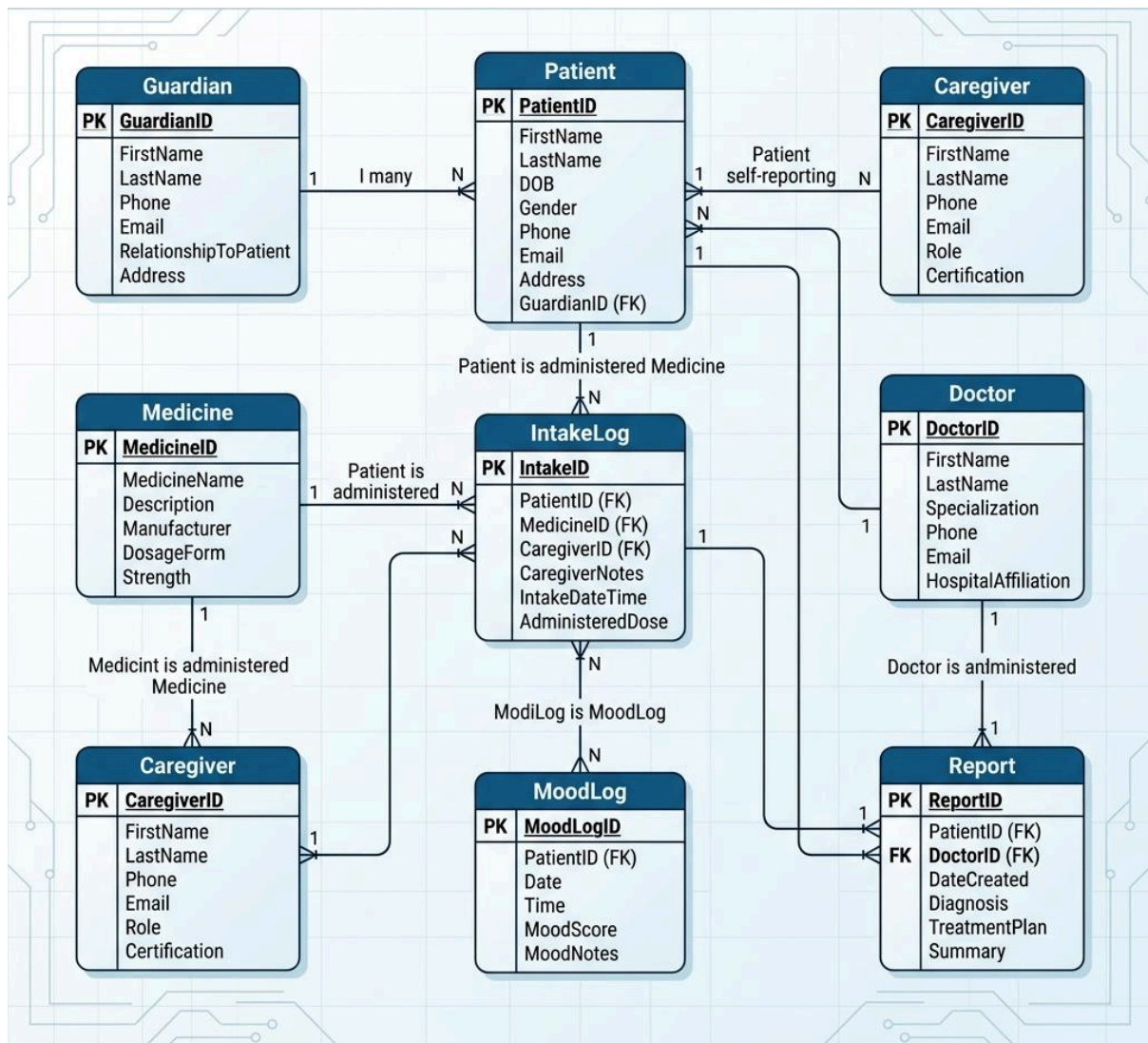
Figure: System Architecture of CarePlus

System Design

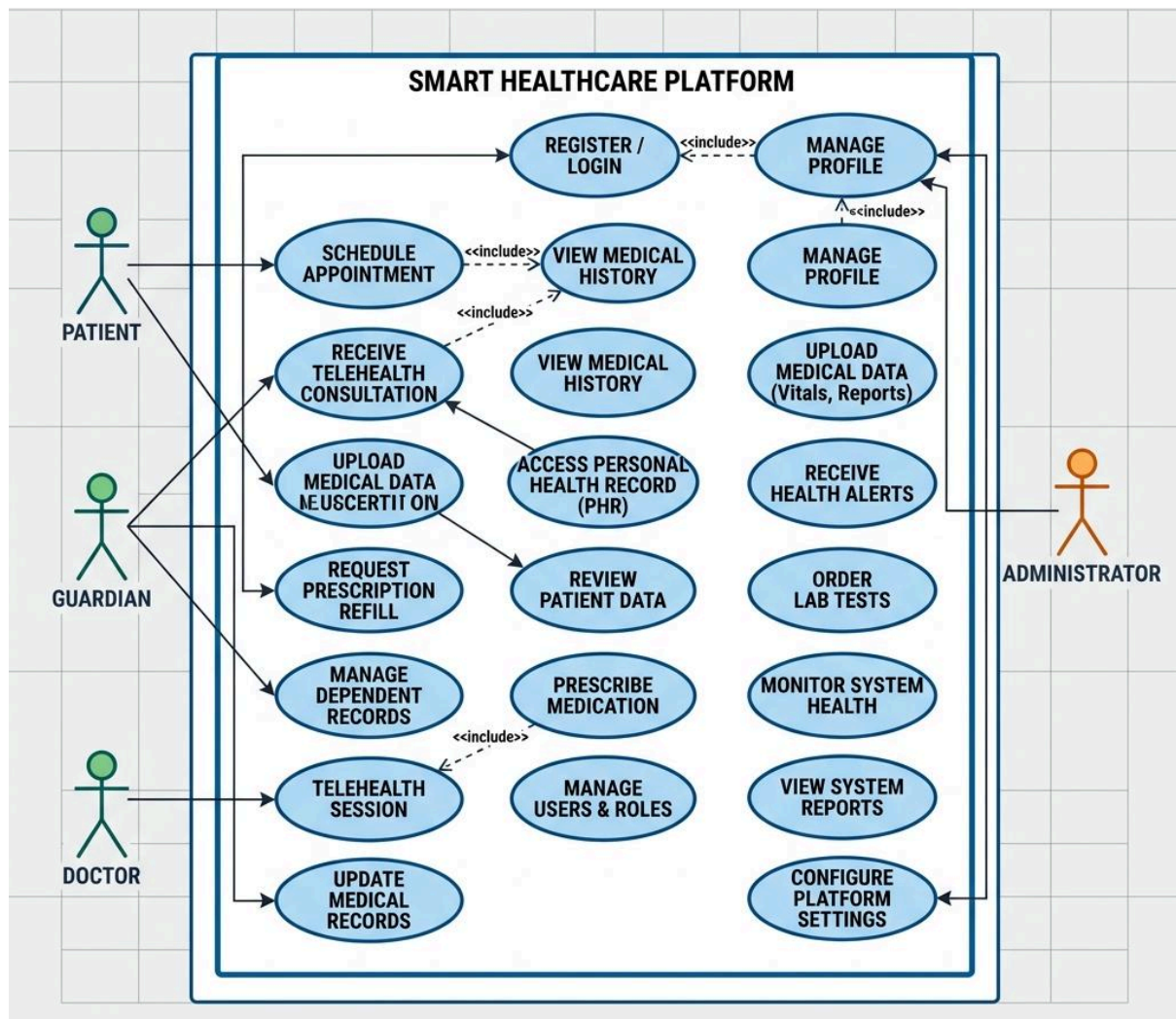
System Architecture



Database Design (ER Diagram)



Use Case Diagram



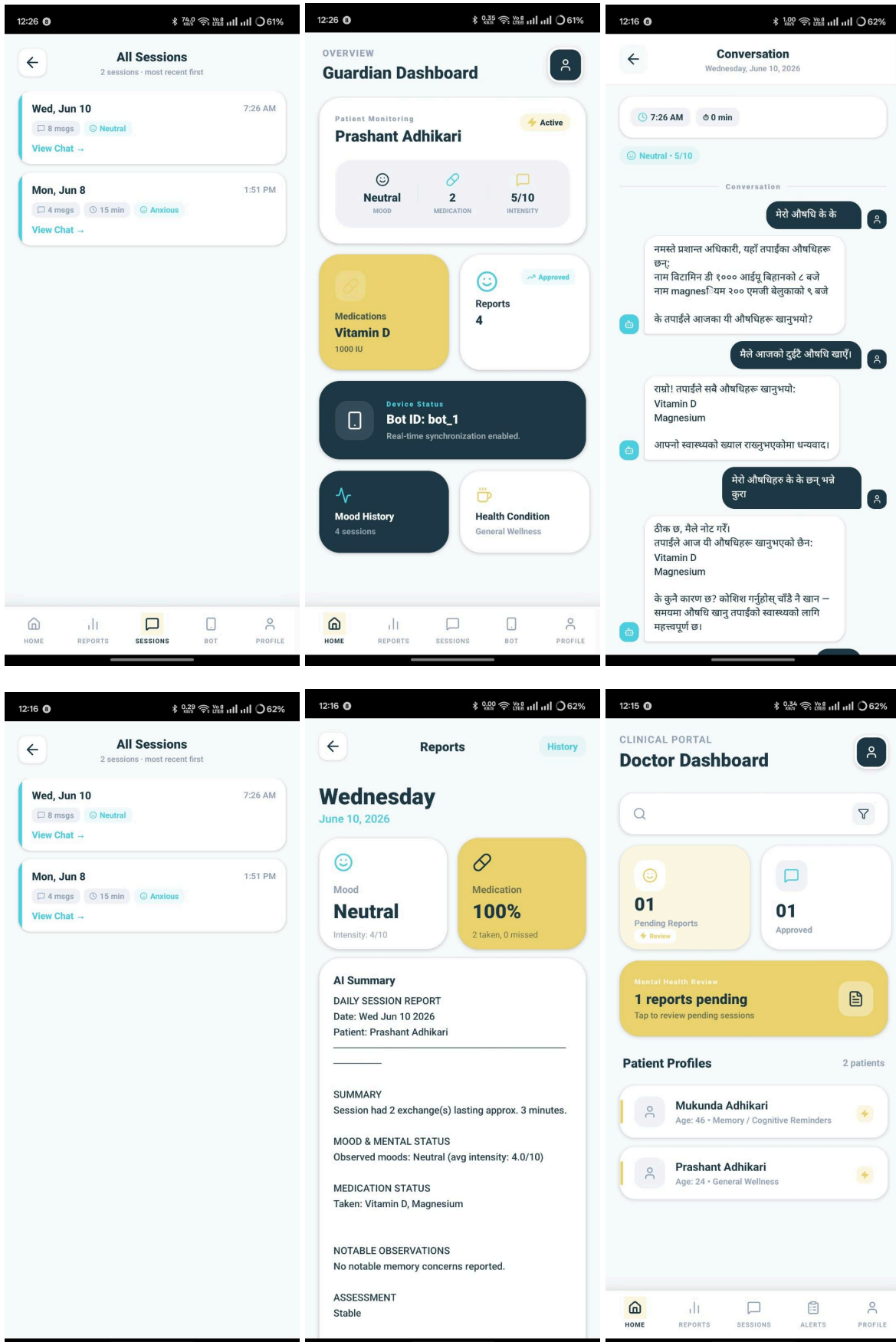
System Flowchart

The core interaction flow proceeds as follows: user speech input, Whisper speech-to-text conversion, FastAPI orchestration, LLM decision engine, action execution (reply, medicine update, object lookup, mood log, or report), text-to-speech output, and dashboard update with data persistence.



Wireframes / UI Design

Doctors, guardians: role based dashboards displaying each patient's mood score, loneliness, psycho-sentiment analysis summary, and adherence to medications.



AI Integration

Question	Answer
Where is AI used?	AI powers voice interaction, emotional analysis, healthcare responses, and personalized reminders.
Why is AI required?	AI enables natural, Nepali-first communication, accessibility for elderly users, and intelligent healthcare coordination.
What value does AI add?	It allows the system to understand free-form spoken requests, infer intent, and generate contextually appropriate responses rather than relying on rigid menus.
What happens if AI is removed?	The system could still provide basic scheduled reminders and static dashboards, but would lose conversational healthcare assistance and emotional trend analysis.

Development Process

The team followed an agile development process with weekly sprint planning and iterative testing. Hardware and software modules were developed in parallel, with responsibilities distributed across the four core project areas.

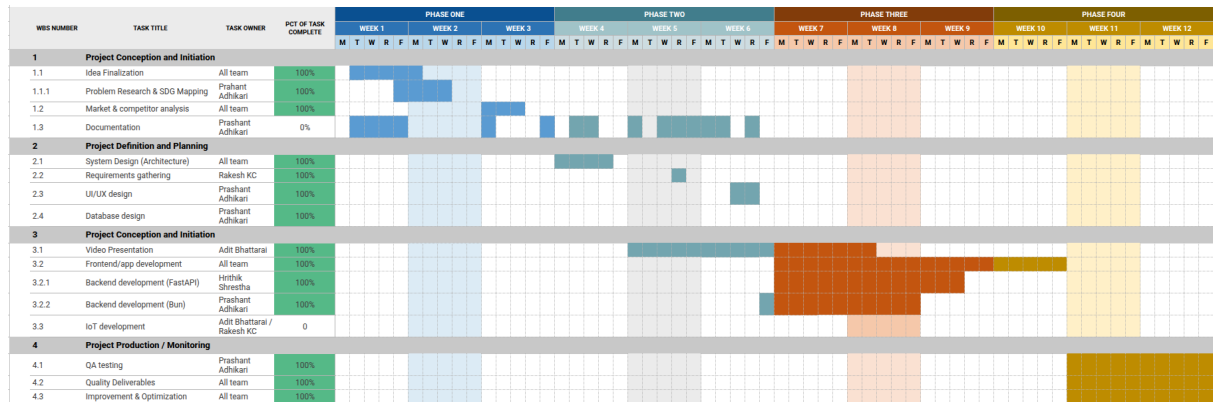
Team Responsibilities

Name	Role	Key Responsibilities
Prashant Adhikari	Core communication & AI orchestration	Speech-to-text/text-to-speech integration, server routing, interaction flow
Adit Bhattarai	Emotional analytics & automated reporting	Mood tracking, JSON reports, PDF reporting pipeline
Hritik Shrestha	Multi-user ecosystem & hardware integration	Raspberry Pi setup, synchronized dashboards, deployment hardening
Rakesh KC	Medical & functional management	Medicine CRUD, care workflows, object-response logic

Timeline / Milestones

- Milestone 1 — Core Communication & AI Orchestration: Whisper STT/TTS integration and server routing foundation.
- Milestone 2 — Medical & Functional Management: medicine CRUD, intake confirmation, and object query responses.
- Milestone 3 — Emotional Analytics & Automated Reporting: mood tracking, JSON report engine, PDF rendering pipeline.
- Milestone 4 — Multi-User Ecosystem & Hardware Integration: Raspberry Pi deployment hardening and synchronized multi-role dashboards.

Gantt Chart



Testing & Validation

Testing Methods

- Unit testing for AI and API modules.
- Hardware reliability testing on the Raspberry Pi device.
- Voice recognition testing in varied and noisy environments.
- Dashboard synchronization testing across roles.
- User acceptance testing with elderly participants.

User Testing and Feedback

Informal testing sessions were conducted with a small number of pilot participants to observe how comfortably elderly users interacted with the voice assistant and how guardians interpreted the dashboard data.

Bug Fixes and Improvements

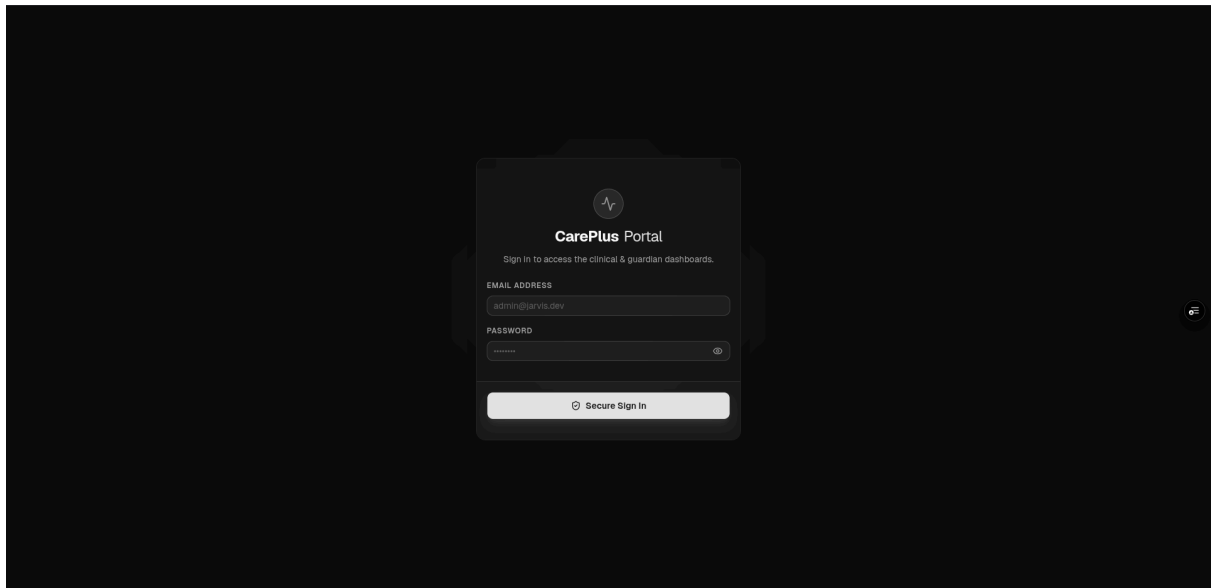
The problems found were reported during testing, such as the occasional problem of the speech not being recognized

in noisy environments and minor synchronization delays between the dashboard panels, and these were resolved by making several adjustments to the audio before processing and by doing a few updates to the backend on an event basis.

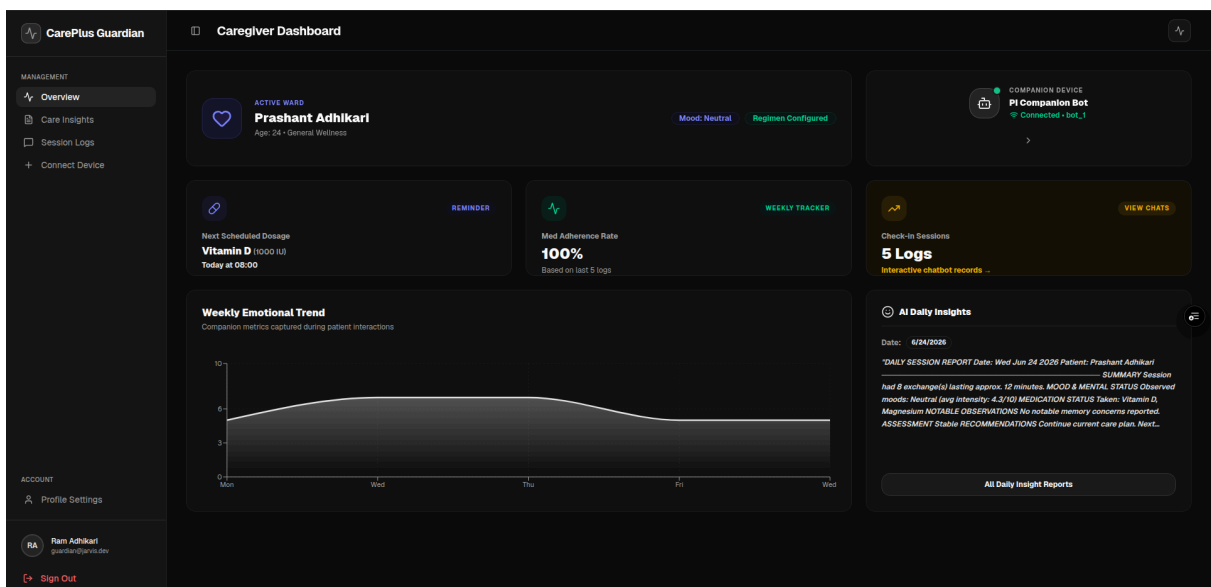
Demonstration of MVP

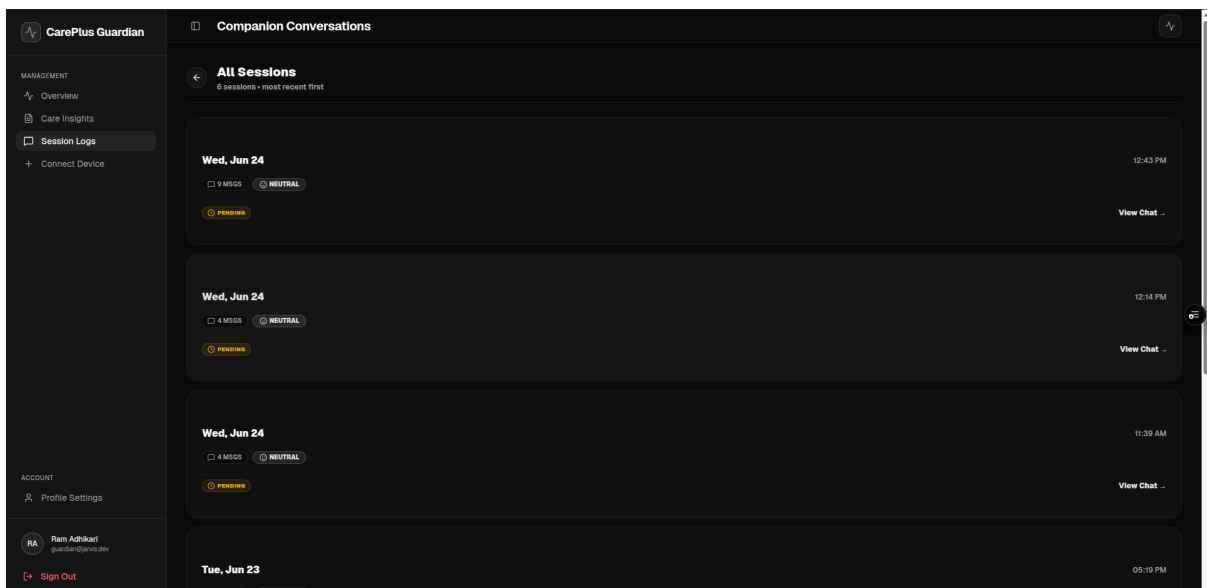
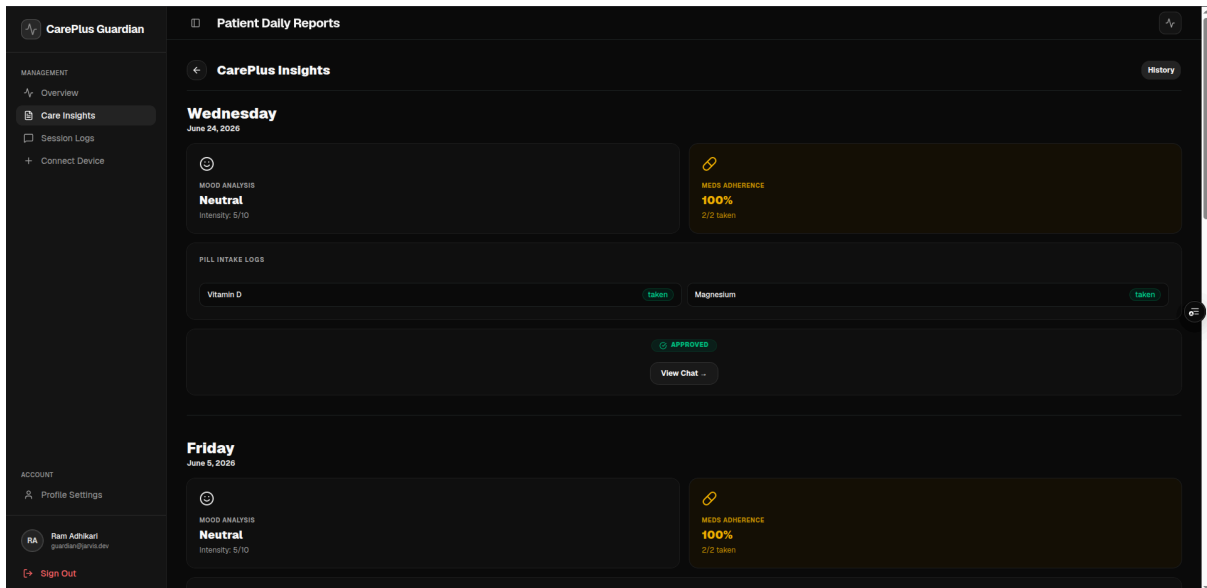
This section presents the working prototype through representative screenshots of the login flow, role-based dashboards, and AI functionality.

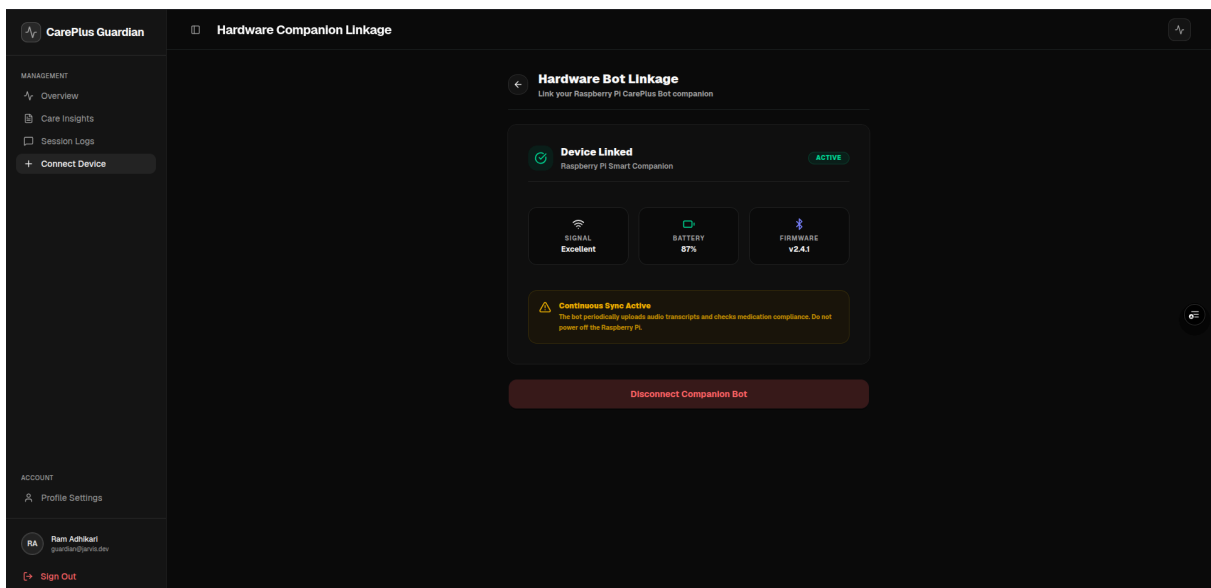
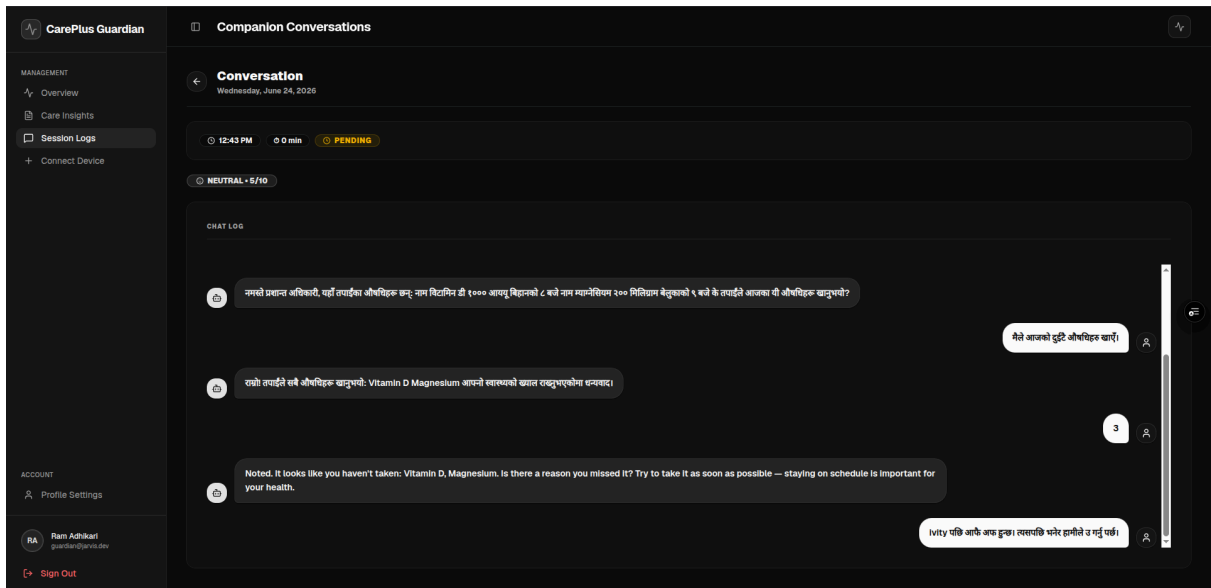
Login



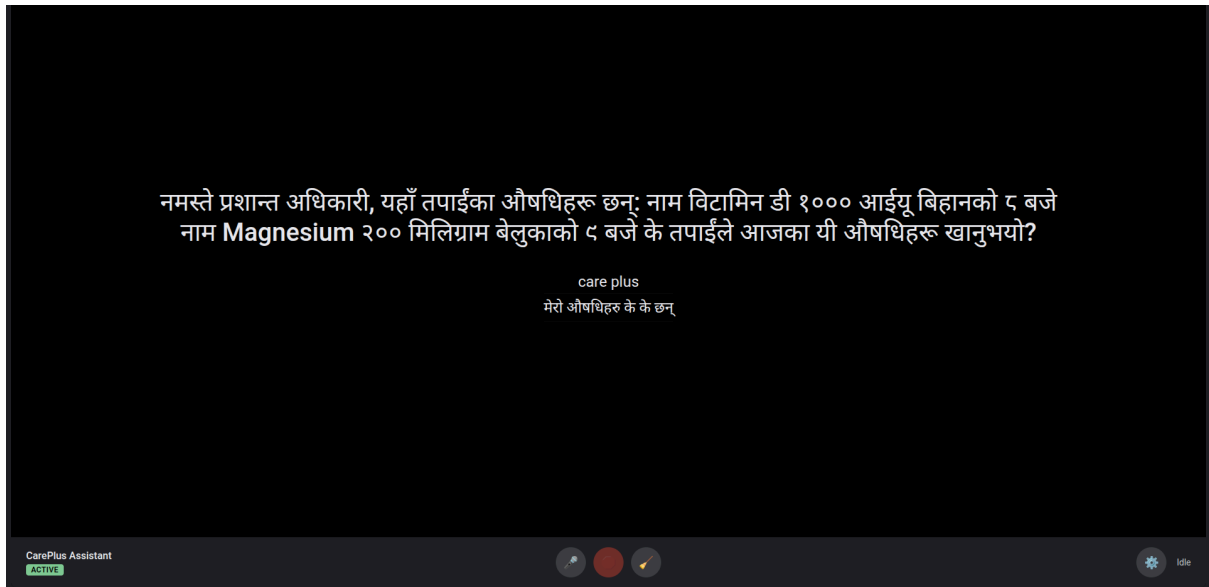
Dashboard







AI features



Final Output

This is CarePlus, an innovative AI powered dementia care robot and loneliness companion, built as the final product for Sunway College Kathmandu's Semester 2 Innovation Project. Our working final build boasts a sturdy 4-module monorepo: on-device AI hardware server (Gemma, RAG and Nepali Whisper STT/TTS), Bun/TypeScript backend, mobile app for guardians, and NextJS dashboard for physicians. The product, in its most basic form, supports contextualized, localized conversational AI to engage dementia patients, identifies their emotional status, and provides them with smart medication-logging questions to provide an exact patient's medical records that are accessible for real time monitoring by both physicians and guardians.

Challenges & Solutions

Technical Challenges

- Speech recognition accuracy degraded in noisy home environments — addressed through audio preprocessing, confidence scoring, and repeat-confirmation prompts.
- Occasional Raspberry Pi hardware or service failure — addressed through watchdog scripts, automatic service restarts, and maintaining spare units.
- Cross-panel dashboard synchronization inconsistency — addressed by adopting an event-driven architecture with clear backend ownership.

Team Challenges

Close coordination among four parallel work streams (AI orchestration, medical logic, analytics, and hardware) was necessary to prevent integration issues, especially any conflicts with shared API contracts and data formats.

Time Management

Others constraints were balancing academic course work with the iterative development and testing cycles of Care+ due to the need to complete one sprint per week and prioritising the critical must have MVP features first.

Learning Experiences

The team developed practical expertise in orchestrating LLM at the local level, and deploying them on edge devices, in local speech processing multilingualism, and designing role-based systems for non-technical users, including senior citizens who are not very tech-savvy.

Project Outcomes

Objectives Achieved

The team successfully delivered a working prototype demonstrating the core Care+ assistance loop, including Nepali-first voice interaction, medicine reminders and intake logging, mood tracking, and synchronized role-based dashboards, in line with the objectives set out in the Introduction.

Customer Value Delivered

Initial informal pilot feedback revealed that spoken interaction was more accessible than text-based applications to the elderly participants and that the dashboard gave guardians greater reassurance when they could see their medication taking and that guardians' moods were changing.

Skills Gained

- Applied experience of speech-to-text and text-to-speech pipelines.
- Orchestration of the backend using FastAPI and synchronization – event-driven.
- Local large language model deployment and prompt-driven action routing.
- User-centred design for old and low digital-literacy people.

Innovation Achieved

Care+ demonstrates a practical, locally hosted alternative to cloud-dependent healthcare applications, combining voice AI, IoT, and multi-role coordination in a way that has not been widely implemented for the Nepali home-care context.

Future Scope

With an additional six months of development, the team would prioritise the following improvements and expansions:

- Formal clinical validation of mood and emotional trend analysis with healthcare professionals.
- Integration of wearable devices for heart rate, sleep, and movement monitoring.
- Expanded multilingual support covering additional South Asian languages.
- Clinical integration APIs to connect Care+ with hospital information systems.
- Scaling pilot deployment from individual households to community and institutional settings, supported by a more robust cloud synchronization and modular backend infrastructure.

Conclusion

What sets Care+ apart is its ability to adapt these technologies to the context of elderly care in the local language, drawing on Nepal's communities and available resources to deliver a range of specialised services that integrate voice interaction with different caregiving tasks and roles. By combining local AI processing, IoT hardware, Nepali-first voice interaction, and role-based caregiver coordination within a single privacy-focused system, Care+ fills a real and unmet need in home-based elderly care in Nepal. The project illustrates that a viable and culturally relevant health companion can be created within an academic time frame and offer a clear pathway to pilot implementation and growth.

Through this Innovation Project, Team Git Force has validated the core assistance loop of Care+, established a realistic business and go-to-market strategy, and identified a concrete roadmap of future improvements. The team thinks that Care+ could make a meaningful contribution towards improving medication adherence, coordinating caregivers and monitoring emotional wellbeing of elderly people living at home in Nepal, and eventually in similar contexts in South Asia.

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